Zero-Shot vs Few-Shot Prompting

A Comparative Evaluation Across ChatGPT, Claude, and Gemini

GENERATIVE AI AND PROMPT ENGINEERING INTERNSHIP

TASK 1

1. Introduction

Prompt Engineering is the practice of designing and refining the instructions given to a large language model (LLM) so that it produces accurate, relevant, and consistent outputs. Because LLMs generate responses based on the wording, structure, and context of the input prompt, the way a task is phrased has a direct impact on the quality of the result. Effective prompt engineering therefore involves selecting the right level of detail, providing appropriate context, and choosing an instructional format that guides the model toward the desired outcome.

Zero-Shot Prompting refers to a technique in which the model is asked to perform a task without being shown any prior examples of correct input-output pairs. The model relies solely on its pre-trained knowledge and the instructions provided in the prompt to infer the expected behavior.

Few-Shot Prompting, in contrast, provides the model with a small number of labeled examples before presenting the actual task. These examples act as a reference pattern, helping the model better understand the expected format, tone, and classification logic before it processes the real input.

The purpose of this experiment is to compare Zero-Shot and Few-Shot Prompting techniques across three major LLM platforms — ChatGPT, Claude, and Gemini — on a practical text classification task, and to evaluate whether providing examples improves the consistency and accuracy of the models' outputs.

2. Task Objective

The objective of this task is to classify customer support messages into one of three categories:

- Complaint
- Question

- Praise

The same set of ten customer support messages was submitted to each of the three LLMs — ChatGPT, Claude, and Gemini — first using a Zero-Shot prompt and then using a Few-Shot prompt. The classifications produced under each condition were recorded and compared to evaluate the performance of Zero-Shot versus Few-Shot Prompting across the three platforms.

3. Zero-Shot Prompting

In this section, each model was given a direct instruction to classify the ten customer support messages without any prior examples.

3.1 ChatGPT

Prompt

```
You are a customer support classification assistant.

Classify each of the following customer support messages into exactly one of these categories:

- Complaint
- Question
- Praise

Rules:
- Assign only one category per message.
- Do not provide explanations.
- Present the results in a markdown table with columns: Message Number | Category.

Messages:

1. My package arrived damaged and I need a replacement.
2. When will my order be delivered?
3. Your support team was incredibly helpful today.
4. The mobile app crashes every time I try to log in.
5. How can I change my account password?
6. I absolutely love the new design!
7. I was charged twice for the same purchase.
```

```
8. Can I cancel my subscription before renewal?
9. Thank you for the outstanding customer service.
10. The website is extremely slow and difficult to use.
```

Output

Message Number	Category
1	Complaint
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question
9	Praise
10	Complaint

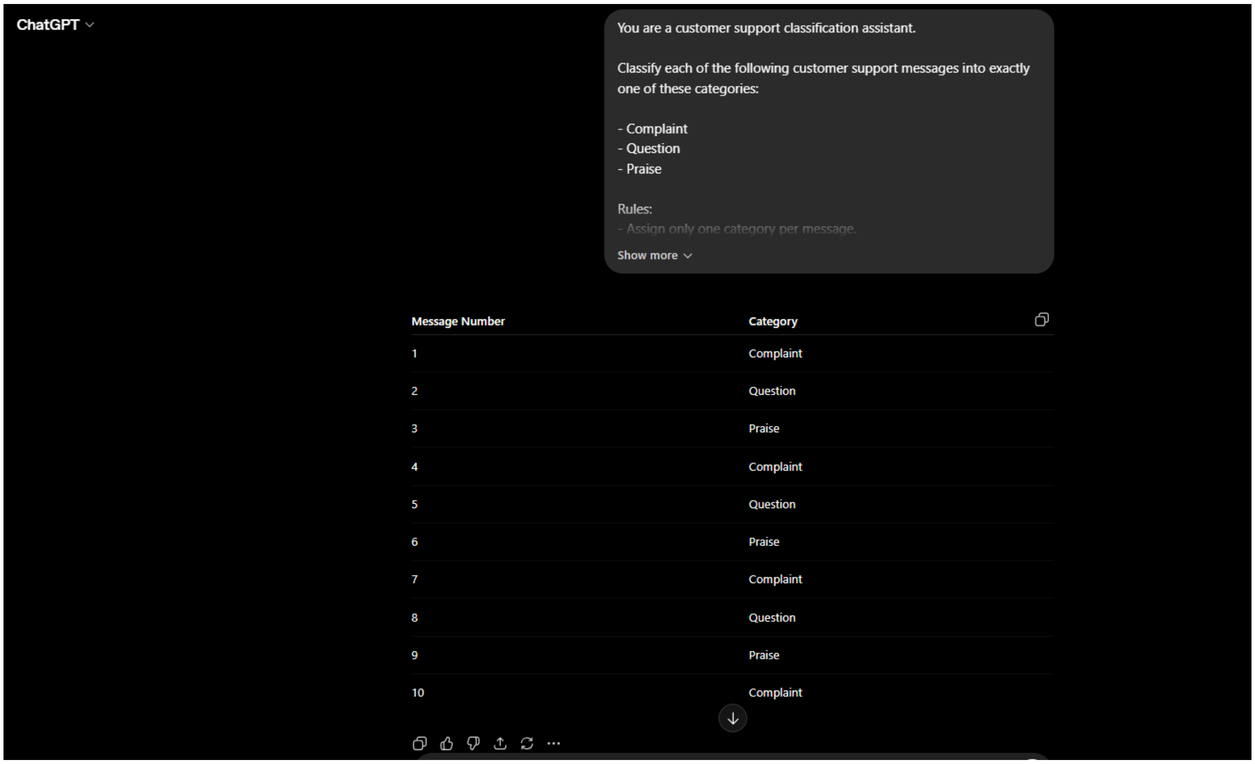

Figure — ChatGPT Zero-Shot Screenshot

3.2 Claude

Prompt

```
Act as a customer service analyst.

Review each customer message and classify it into one of the following categories:

• Complaint
• Question
• Praise

Requirements:
- Select only one category per message.
- Do not include any reasoning or additional comments.
- Return the output as a numbered table with two columns: Message Number and Classification.

Customer Messages:

1. My package arrived damaged and I need a replacement.
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```
2. When will my order be delivered?
3. Your support team was incredibly helpful today.
4. The mobile app crashes every time I try to log in.
5. How can I change my account password?
6. I absolutely love the new design!
7. I was charged twice for the same purchase.
8. Can I cancel my subscription before renewal?
9. Thank you for the outstanding customer service.
10. The website is extremely slow and difficult to use.
```

Output

Message Number	Classification
1	Complaint
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question
9	Praise
10	Complaint

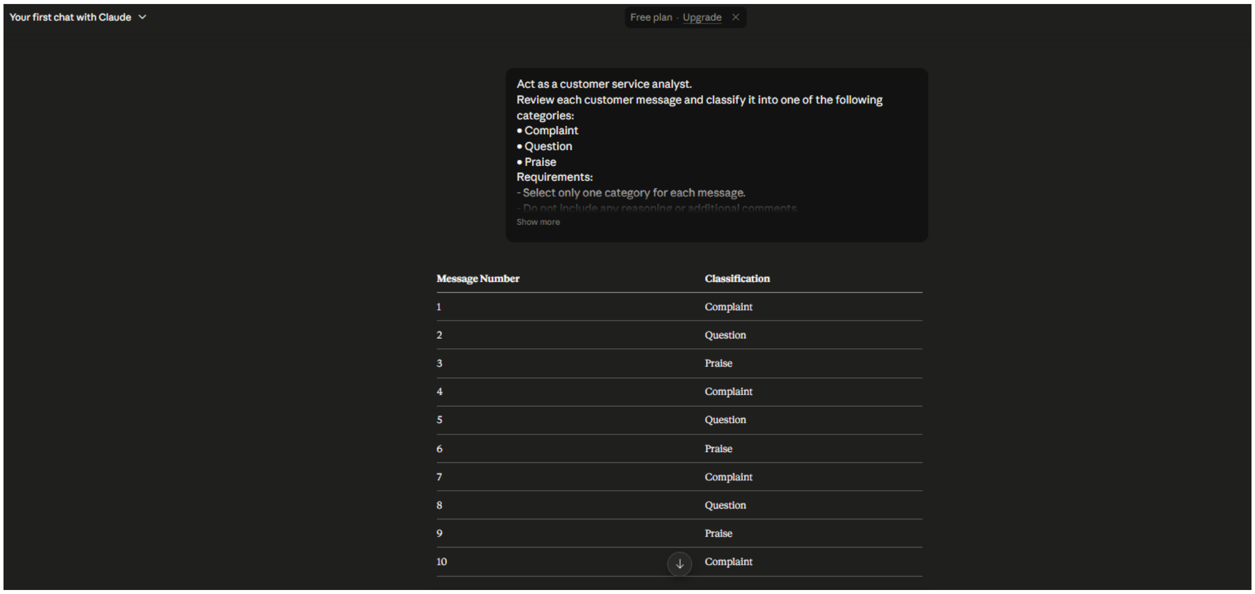

Figure — Claude Zero-Shot Screenshot

3.3 Gemini

Prompt

```
You are an AI assistant specializing in customer support analysis.

For each message below, determine whether it represents:

- Complaint
- Question
- Praise

Instructions:
- Use only one label per message.
- Do not explain your decisions.
- Display the final answer in a table with the columns: Message Number and Category.

Messages:

1. My package arrived damaged and I need a replacement.
2. When will my order be delivered?
3. Your support team was incredibly helpful today.
4. The mobile app crashes every time I try to log in.
5. How can I change my account password?
6. I absolutely love the new design!
7. I was charged twice for the same purchase.
```

```
8. Can I cancel my subscription before renewal?
9. Thank you for the outstanding customer service.
10. The website is extremely slow and difficult to use.
```

Output

Message Number	Category
1	Complaint
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question
9	Praise
10	Complaint

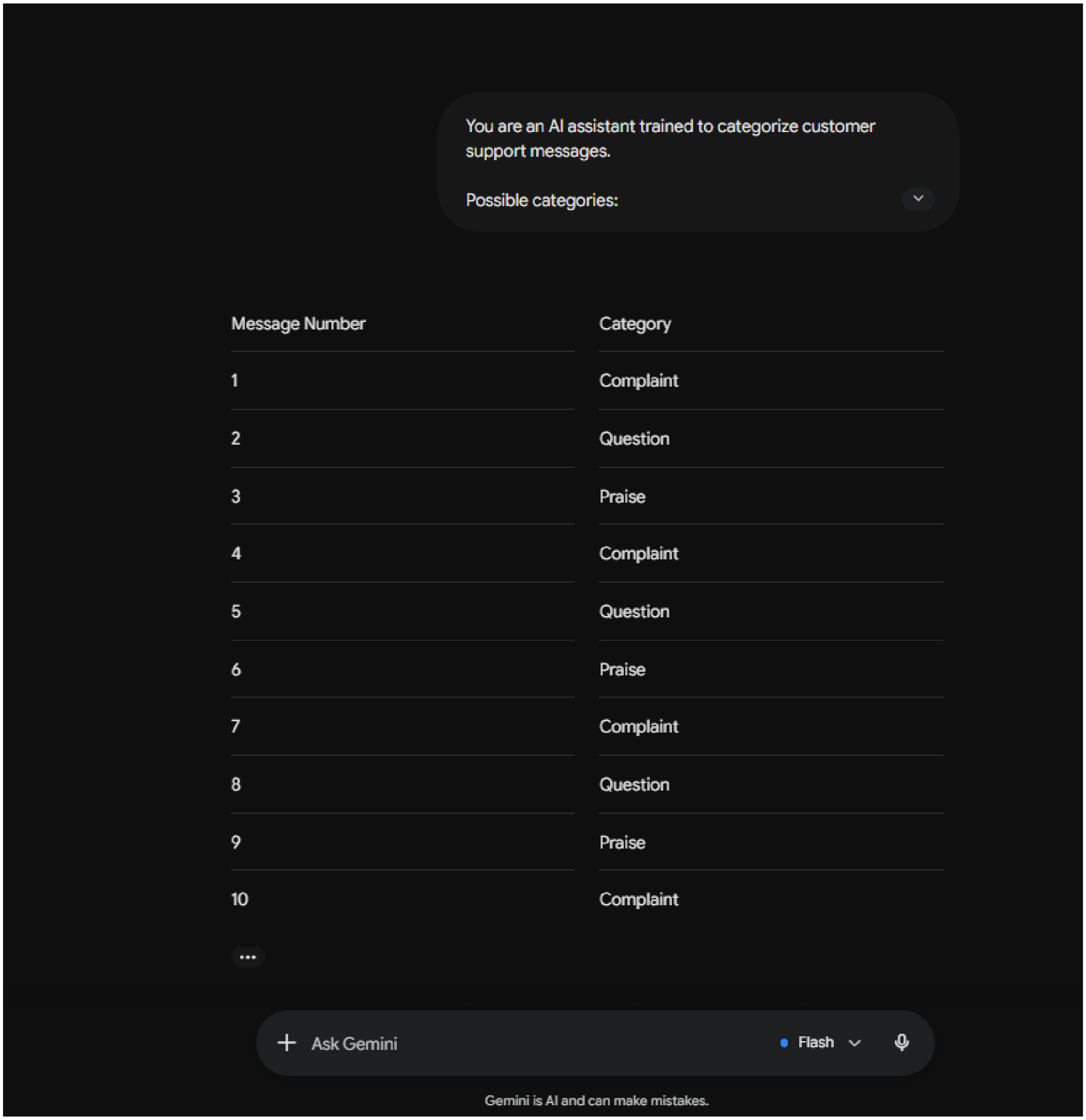

Figure — Gemini Zero-Shot Screenshot

4. Few-Shot Prompting

In this section, each model was given the same classification task, preceded by three labeled examples demonstrating the expected category for a sample message.

4.1 ChatGPT

Prompt

```
You are a customer support classification assistant.

Classify each customer message into one of these categories:

- Complaint
- Question
- Praise

Examples:

Message: My internet keeps disconnecting every evening.
Category: Complaint

Message: What are your customer support hours?
Category: Question

Message: Excellent service! I really appreciate your quick response.
Category: Praise

Now classify the following messages.

1. My package arrived damaged and I need a replacement.
2. When will my order be delivered?
3. Your support team was incredibly helpful today.
4. The mobile app crashes every time I try to log in.
5. How can I change my account password?
6. I absolutely love the new design!
7. I was charged twice for the same purchase.
8. Can I cancel my subscription before renewal?
9. Thank you for the outstanding customer service.
10. The website is extremely slow and difficult to use.

Return only a markdown table with Message Number and Category.
```

Output

Message Number	Category
1	Complaint

Message Number	Category
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question
9	Praise
10	Complaint

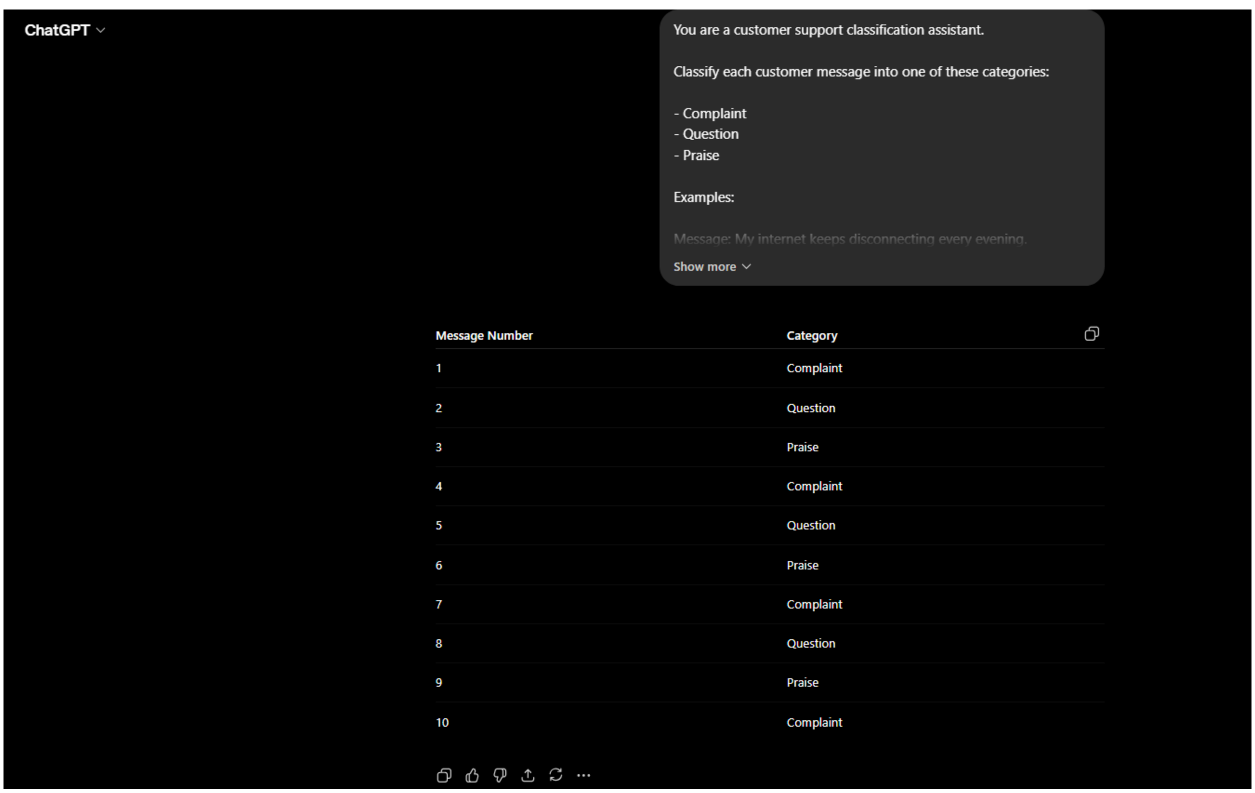

Figure — ChatGPT Few-Shot Screenshot

4.2 Claude

Prompt

```
Act as a customer support analyst.

Your task is to classify each message into one of these categories:

• Complaint
• Question
• Praise

Reference Examples:

Message: My order was delivered to the wrong address.
Classification: Complaint

Message: Is international shipping available?
Classification: Question

Message: I'm very impressed with your excellent support team.
Classification: Praise

Now classify the following customer messages.

1. My package arrived damaged and I need a replacement.
2. When will my order be delivered?
3. Your support team was incredibly helpful today.
4. The mobile app crashes every time I try to log in.
5. How can I change my account password?
6. I absolutely love the new design!
7. I was charged twice for the same purchase.
8. Can I cancel my subscription before renewal?
9. Thank you for the outstanding customer service.
10. The website is extremely slow and difficult to use.

Return your answer as a two-column table (Message Number | Classification).
```

Output

Message Number	Classification
1	Complaint
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question
9	Praise
10	Complaint

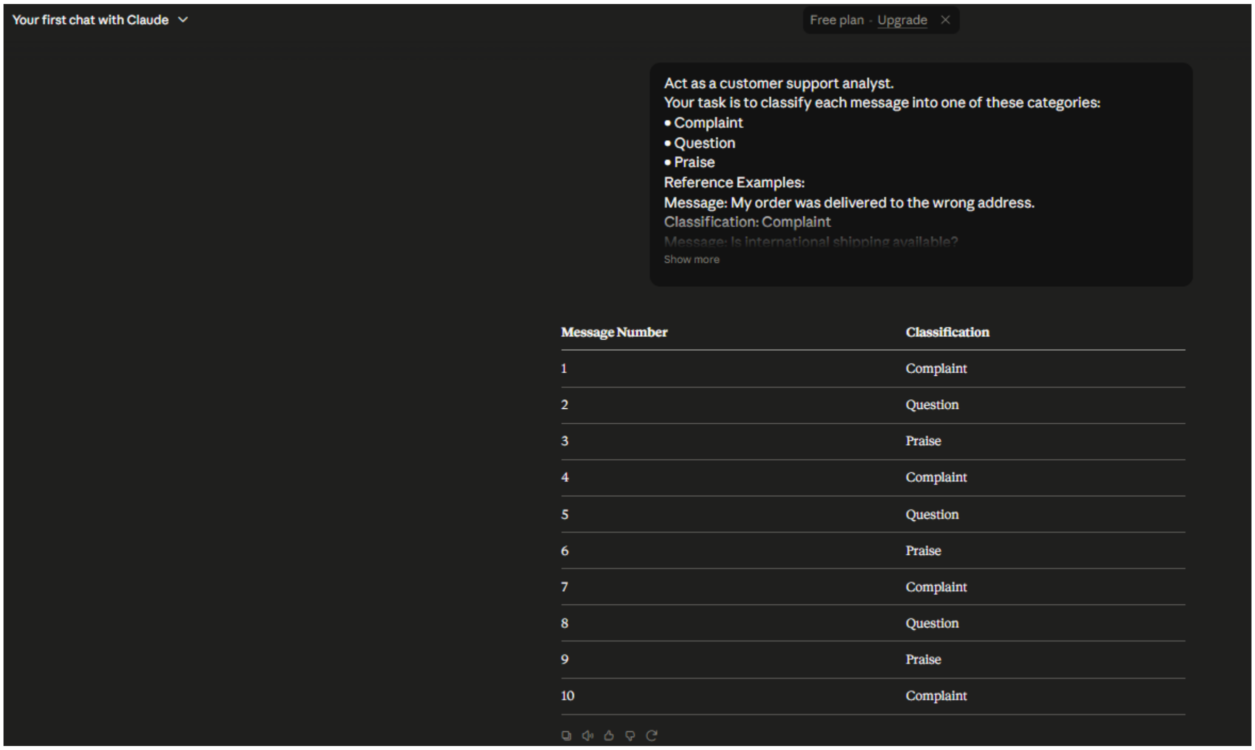

Figure — Claude Few-Shot Screenshot

4.3 Gemini

Prompt

```
You are an AI assistant trained to categorize customer support messages.

Possible categories:

- Complaint
- Question
- Praise

Training Examples:

Example 1
Message: My payment failed but the money was deducted.
Category: Complaint

Example 2
Message: How do I track my shipment?
Category: Question
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Example 3
Message: Your service exceeded my expectations.
Category: Praise

Now classify these messages.

1. My package arrived damaged and I need a replacement.
2. When will my order be delivered?
3. Your support team was incredibly helpful today.
4. The mobile app crashes every time I try to log in.
5. How can I change my account password?
6. I absolutely love the new design!
7. I was charged twice for the same purchase.
8. Can I cancel my subscription before renewal?
9. Thank you for the outstanding customer service.
10. The website is extremely slow and difficult to use.

Output only a table with Message Number and Category.
```

Output

Message Number	Category
1	Complaint
2	Question
3	Praise
4	Complaint
5	Question
6	Praise
7	Complaint
8	Question

Message Number	Category
9	Praise
10	Complaint

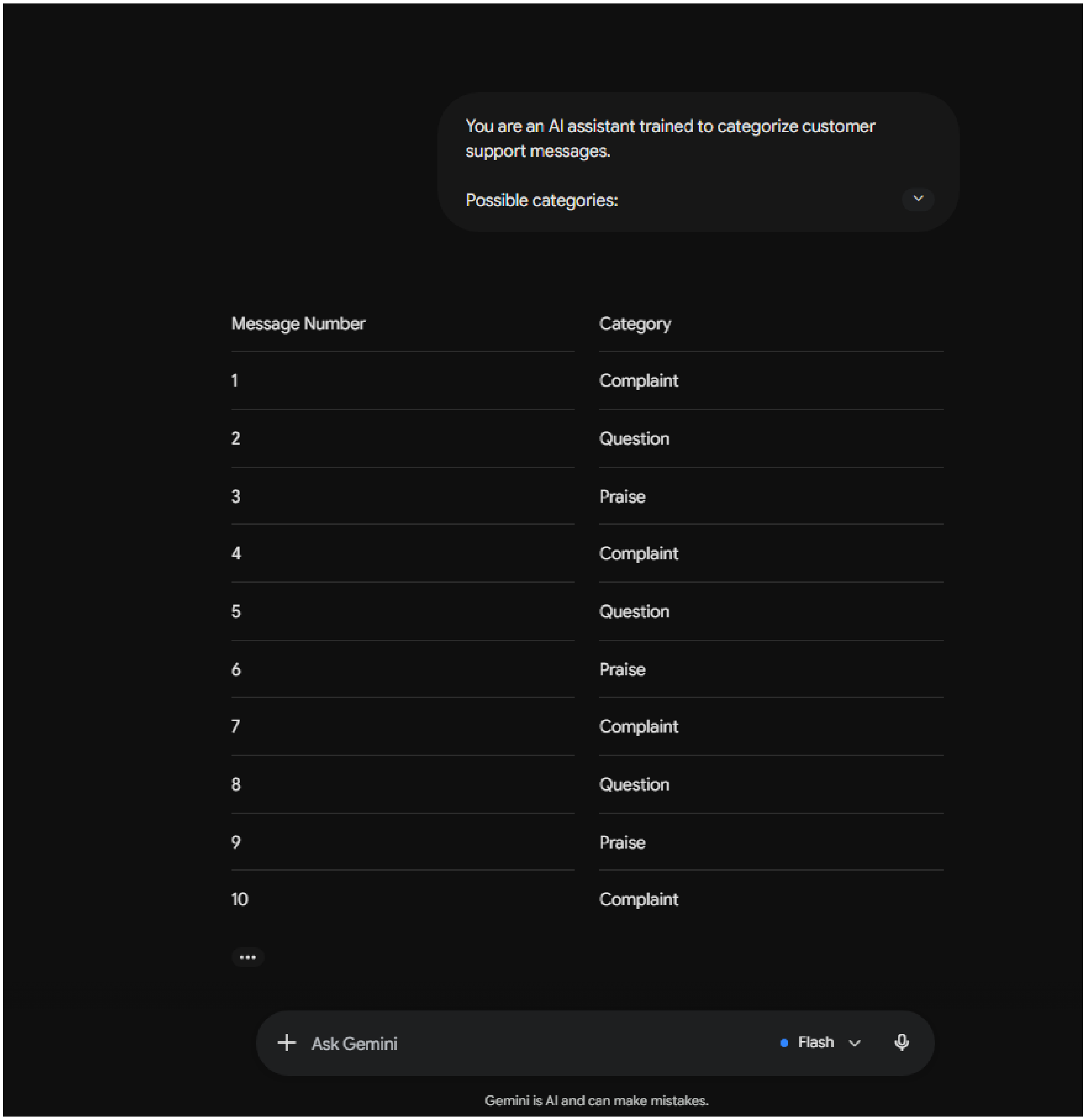

Figure — Gemini Few-Shot Screenshot

5. Results

All three LLMs — ChatGPT, Claude, and Gemini — correctly classified all ten customer support messages under both the Zero-Shot and Few-Shot prompting conditions. Each model consistently assigned the appropriate category (Complaint, Question, or Praise) to every message, and no misclassifications were observed in either approach. The full classification outputs for each model and prompting method are presented in Section 3 and Section 4 above.

6. Accuracy Comparison Table

LLM	Zero-Shot	Few-Shot	Observation
ChatGPT	10/10 (100%)	10/10 (100%)	Correctly classified every message in both approaches.
Claude	10/10 (100%)	10/10 (100%)	Produced completely accurate classifications in both cases.
Gemini	10/10 (100%)	10/10 (100%)	Achieved perfect accuracy using both prompting methods.

7. Analysis

The results of this experiment show that all three language models — ChatGPT, Claude, and Gemini — were able to achieve perfect classification accuracy on the customer support message dataset, regardless of whether a Zero-Shot or Few-Shot prompting strategy was used. This suggests that for a relatively simple, well-defined, three-class classification task with clearly worded messages, the underlying language understanding capabilities of current-generation LLMs are strong enough to succeed without requiring demonstration examples.

Despite the identical accuracy scores, the Zero-Shot and Few-Shot prompts differed meaningfully in structure. The Zero-Shot prompts relied entirely on explicit rule-based instructions — defining the categories, specifying the output format, and prohibiting explanatory text — to guide the models' behavior. The Few-Shot prompts, on the other hand, supplemented these instructions with three labeled example messages, giving the models a concrete pattern to follow for tone, format, and classification logic before the actual task began.

Because the messages used in this evaluation contained fairly unambiguous linguistic cues — for example, phrases indicating dissatisfaction ('arrived damaged', 'charged twice'), direct requests for information ('when will', 'how can I'), and clear expressions of gratitude or satisfaction ('outstanding customer service', 'love the new design') — all three models were likely able to rely on strong pattern recognition from their pre-training alone. In tasks with more ambiguous, sarcastic, or mixed-sentiment messages, the benefit of Few-Shot examples in anchoring the model's interpretation would likely become more pronounced and could result in a measurable accuracy gap between the two prompting strategies.

Overall, this comparison highlights that prompt design still plays an important role in ensuring consistent, well-formatted output — even when the underlying classification task is easily handled by the model — and that the marginal benefit of Few-Shot examples is task-dependent rather than universal.

8. Conclusion

This experiment compared Zero-Shot and Few-Shot Prompting for customer support message classification across ChatGPT, Claude, and Gemini. All three models achieved 100% accuracy under both prompting strategies, indicating that for this particular task, well-structured instructions alone were sufficient to produce reliable results. While Few-Shot Prompting did not yield a measurable accuracy improvement in this specific evaluation, it remains a valuable technique for guiding output format and consistency, and is expected to offer greater benefit on more complex or ambiguous classification tasks. Future work could extend this evaluation to messages with mixed sentiment or domain-specific vocabulary to better assess the comparative strengths of Zero-Shot and Few-Shot approaches.